

Three Fermion Generations as CP/T-Asymmetric Identity Transformation: The CKM Lower Bound, Irreversible Pruning, and the Minimal Flavor Architecture of Finite Distinction Systems

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The Standard Model contains three generations of chiral fermions with repeated gauge quantum numbers and different masses. Chiral gauge assignments are essential: before electroweak symmetry breaking they forbid bare gauge-invariant Dirac mass terms, so sequential generations appear as protected weak-flavor distinction sectors rather than vector-like pairs. The Standard Model takes this number as empirical input. X2 studies a narrower algebraic question: what dimensionality is required for a weak-sector identity-changing channel to carry a rephasing-invariant CP/T orientation? In a CKM-type $N \times N$ unitary charged-current mixing matrix, the number of physical complex phases that cannot be removed by quark-field rephasings is $(N-1)(N-2)/2$. Hence $N=1$ and $N=2$ cannot support an irreducible CKM CP phase, while $N=3$ is the first possible case. X2 places this Kobayashi-Maskawa fact inside the FDS/DT physical bridge chain

identity update $\rightarrow$ T/CP -orientation $\rightarrow$ irreducible CKM phase $\rightarrow N_{\text{CKM}} \geq 3$.

The hard result is a conditional lower bound: if weak-sector identity transformation requires a microscopic CP/T-asymmetric carrier, and if the Standard Model charged-current sector is that carrier, then the CKM-participating flavor dimension must satisfy $N_{\text{CKM}} \geq 3$. This is a necessary dimensional lower bound, not a sufficient derivation of observed CP violation; nonzero mixing, nondegenerate masses, and a nonzero Jarlskog invariant are additional physical requirements. The stronger claim that the observed theory realizes exactly three sequential chiral generations, rather than more, is separated as a higher-risk finite-system minimality bridge. Deterministic algebraic demonstrations illustrate CKM phase counting, quark rephasing, rephasing-invariant orientation, hard/soft claim separation, and a toy flavor-cost functional.

Reader contract. X2 is a conditional physical-bridge paper. It does not derive the Standard Model gauge group, the fermion mass spectrum, Yukawa couplings, CKM angles, the numerical CKM phase, neutrino masses, or the observed baryon asymmetry. It does not claim that all irreversibility requires CKM CP violation. The CP/T-asymmetry bridge applies to microscopic identity-changing weak-sector transformations, not to all macroscopic thermodynamic irreversibility, Landauer erasure, decoherence, or open-system dissipation. The CKM phase-counting result supplies a necessary dimensional lower bound, not a sufficient derivation of observed CP violation. The hard claim is $N_{\text{CKM}} \geq 3$ for CKM-participating weak flavor. The claim that the realized sequential chiral generation count is exactly three is a separate minimality bridge with a different epistemic status.

Keywords

Finite Distinction Systems; Distinction Theory; fermion generations; CKM matrix; CP violation; T violation; CPT theorem; Kobayashi-Maskawa phase; Jarlskog invariant; identity transformation; weak interaction; flavor physics; PMNS matrix; fourth generation; Standard Model.

INTRODUCTION

The flavor-dimensionality problem

The Standard Model contains three repeated families of chiral fermions. In the quark sector these are

$$(u, d), \quad (c, s), \quad (t, b), \quad (1)$$

with corresponding charged leptons and neutrinos. The gauge quantum numbers repeat, while masses and mixings differ. The Standard Model describes these patterns with Yukawa matrices and charged-current mixing, but it does not derive why the weak flavor architecture has this dimensionality rather than one, two, four, or more [7].

X2 focuses on one sharp subproblem. A CKM-type weak charged-current sector can carry a physical CP-violating phase only when its flavor dimension is at least

Claim-status summary

Table I separates the algebraic lower bound from the FDS/DT bridge and the higher-risk upper-bound claim.

TABLE I. Central X2 claims, status, and failure or demotion conditions.

Claim	Status	What would weaken or falsify it
CKM-type CP violation requires $N_{\text{CKM}} \geq 3$.	Standard Model algebra / hard hook	A two-generation CKM-like unitary charged-current theory with a physical irreducible CP phase.
Weak-sector identity transformation requires a rephasing-invariant CP/T orientation.	FDS/DT physical bridge	A complete physical account of the audited microscopic weak identity-changing channel with no CP/T orientation, no compensating asymmetry, and no hidden orientation variable.
The weak charged current is the Standard Model identity-transformation carrier.	Interpretive bridge	Identity-changing weak-sector operations are realized by a different physical carrier outside the CKM charged current.
$N_{\text{CKM}} \geq 3$ follows from the X2 chain.	Conditional theorem	Failure of any premise: the CP/T-orientation bridge, CPT assumptions, the weak-carrier mapping, or CKM phase counting.
Exactly three sequential chiral generations follow from minimality.	Higher-risk upper-bound bridge	Discovery of a fourth sequential chiral generation with Standard Model gauge charges and weak charged-current participation, or a successful $N > 3$ flavor architecture that reduces the relevant cost functional.
Nonzero leptonic Dirac CP phase.	Optional PMNS extension	High-precision oscillation data establish $\delta_{\text{CP}} = 0$ or π under assumptions requiring leptonic CP/T-oriented identity transformation.

three [5, 6]. The new claim is not that this algebra is novel. The new claim is that FDS/DT supplies a reason why a CP/T-oriented identity-changing weak channel is required.

The algebraic hook

For N generations, the number of physical complex phases in a CKM-like unitary mixing matrix is

$$N_{\text{CP}}(N) = \frac{(N-1)(N-2)}{2}. \quad (2)$$

Thus $N = 1$ and $N = 2$ have no physical CKM CP phase, while $N = 3$ has one. This is the hard algebraic hook. X2 builds the conditional chain

weak identity transformation requires CP/T orientation $\Rightarrow$ irreducible CKM phase $\Rightarrow N_{\text{CKM}} \geq 3$ (3)

The variable N_{CKM} denotes the CKM-participating weak flavor dimension. In the Standard Model, N_{CKM} coincides with the number of sequential chiral quark generations.

Hard lower bound versus upper-bound bridge

The lower bound $N_{\text{CKM}} \geq 3$ is algebraic once the bridge assumptions are accepted. The stronger statement that the realized sequential chiral generation count is exactly three is not the same kind of result. Since $N > 3$ also has physical CP phases, CKM algebra alone does not forbid more generations. The upper-bound claim

must therefore be formulated as a finite-system minimality bridge: $N = 3$ is the first sufficient point, while $N > 3$ adds overhead that must justify itself.

FDS/DT BACKGROUND

Weak identity transformation

Let $\mathcal{H}_{\text{id}}$ denote the Hilbert subspace spanned by weak flavor identity labels. In the Standard Model charged-current sector, a down-type mass eigenstate can transform into a superposition of up-type mass eigenstates through the CKM matrix:

$$\mathcal{T}_{\text{id}} : |d_j\rangle \mapsto \sum_i V_{ij} |u_i\rangle. \quad (4)$$

Equivalently, the charged-current interaction is

$$J_{\text{cc}}^\mu = \bar{u}_i \gamma^\mu (1 - \gamma^5) V_{ij} d_j. \quad (5)$$

X2 calls Eq. (4) a weak identity transformation: it changes flavor identity sector while preserving a physical transition history. This is not a derivation of the weak interaction. It identifies the existing weak charged-current sector as the object-level Standard Model carrier of identity-changing transformations.

CP/T-oriented identity transformation

A weak identity transformation is nontrivial when $\mathcal{T}_{\text{id}}$ is not simultaneously diagonal in the relevant mass ba-

sis. It is CP/T-oriented when it contains rephasing-invariant orientation data. The standard invariant test is a Jarlskog-type quartet

$$J_{ijkl} = \text{Im} (V_{ij} V_{kl} V_{il}^* V_{kj}^*). \quad (6)$$

Observable CKM CP violation additionally requires nonzero mixing and nondegenerate quark masses. Thus phase counting is necessary, not sufficient. It tells us when a physical orientation channel can exist; it does not by itself derive the measured CP-violating observables.

Definition 1 (CP/T-oriented weak identity channel). *A CKM-type weak identity channel is CP/T-oriented when there exists at least one rephasing-invariant quartet with $J_{ijkl} \neq 0$, under a nondegenerate mass structure and nonzero mixing configuration.*

Why ordinary dissipative irreversibility is not enough for X2

X2 does not claim that all irreversibility requires CP/T violation. Macroscopic thermodynamic irreversibility, Landauer erasure, decoherence, and open-system dissipation can occur without CKM CP violation. Those processes can obtain an arrow from coarse-graining, environmental boundary conditions, low-entropy initial conditions, or reservoir structure.

The X2 bridge is narrower. It concerns an intrinsic microscopic orientation in the weak-sector identity-changing channel itself. A reversible flavor transformation can mix labels; a CP/T-oriented identity transformation carries rephasing-invariant data that distinguishes a transformation history from its CP/T-reversed counterpart. A removable phase is a convention. A nonzero Jarlskog invariant is physical orientation.

Assumption 1 (Weak-sector CP/T orientation bridge). *For the audited microscopic weak identity-changing channel, irreversible identity transformation requires at least one rephasing-invariant CP/T orientation. This assumption is not a theorem of Standard Model algebra. It is the FDS/DT physical bridge that connects finite-system identity update to the Standard Model CKM algebra.*

Why X2 precedes the four-force mapping

X2 does not need a complete functional taxonomy of all four forces. It only uses the weak charged current as the Standard Model sector in which flavor identity transformations occur. A later four-force paper may use X2 as input: X2 establishes why the weak identity-transformation sector must have enough flavor dimensionality to carry a CP/T orientation.

TABLE II. CKM-type phase counting for N generations.

N	mixing angles	physical CP phases
1	0	0
2	1	0
3	3	1
4	6	3
5	10	6

CKM PHASE COUNTING

Unitary mixing and rephasing

Let $V \in U(N)$ be an $N \times N$ unitary charged-current mixing matrix. A general $U(N)$ matrix has N^2 real parameters. These may be counted as

$$N_{\text{ang}} = \frac{N(N-1)}{2} \quad (7)$$

real mixing angles and

$$N_{\text{phase,raw}} = \frac{N(N+1)}{2} \quad (8)$$

raw phases.

Quark fields can be redefined by

$$u_i \rightarrow e^{i\alpha_i} u_i, \quad d_j \rightarrow e^{i\beta_j} d_j. \quad (9)$$

There are $2N$ such field phases, but one common overall phase is unobservable. Hence $2N - 1$ phases are removable. The number of physical CP phases is therefore

$$N_{\text{CP}}(N) = \frac{N(N+1)}{2} - (2N-1) \quad (10)$$

$$= \frac{N^2 - 3N + 2}{2} \quad (11)$$

$$= \frac{(N-1)(N-2)}{2}. \quad (12)$$

Theorem 1 (CKM CP-phase lower bound). *For a CKM-type $N \times N$ unitary charged-current mixing matrix, an irreducible physical complex phase exists if and only if $N \geq 3$.*

Proof. By Eq. (12), the number of physical complex phases is $(N-1)(N-2)/2$. This number is strictly positive if and only if $N \geq 3$. $\square$

Remark 1 (Necessary, not sufficient). *A positive phase count only means that CP/T orientation is algebraically possible. Observable CKM CP violation requires nonzero mixing angles, nondegenerate masses, and a nonzero rephasing-invariant quantity such as Eq. (6). The phase-counting theorem is a dimensional lower bound, not a complete derivation of CKM phenomenology.*

MAIN X2 THEOREM

Theorem 2 (Three-flavor lower bound for CP/T-oriented weak identity transformation). *Assume: (i) the audited weak identity-changing channel requires rephasing-invariant CP/T orientation; (ii) the Standard Model charged-current CKM structure is the relevant carrier; (iii) CPT holds, so microscopic T asymmetry is represented as CP asymmetry under the usual local Lorentz-invariant QFT assumptions; and (iv) CP asymmetry in this sector requires an irreducible CKM phase. Then the CKM-participating weak flavor dimension satisfies*

$$N_{\text{CKM}} \geq 3. \quad (13)$$

In the Standard Model, this lower bound is the lower bound on sequential chiral quark generations participating in the CKM charged current.

Proof. Assumption (i) requires a rephasing-invariant CP/T orientation. Assumption (ii) locates this orientation in the CKM charged-current mixing matrix. Assumption (iii) permits the usual bridge from T asymmetry to CP asymmetry under CPT [8, 9]. Assumption (iv) requires a physical irreducible CKM phase. By the CKM CP-phase lower-bound theorem, such a phase exists only when $N_{\text{CKM}} \geq 3$. $\square$

The theorem should be read exactly as a conditional lower bound. It does not prove the Standard Model gauge group, the measured values of masses, or the measured values of the CKM matrix. It proves that if weak identity transformation needs CKM-type CP/T orientation, two generations are not enough.

EXACTLY THREE? THE MINIMALITY BRIDGE

Algebraic permissibility of $N > 3$

CKM algebra permits $N > 3$. For $N = 4$, Eq. (12) gives three physical CP phases. Therefore the step from $N_{\text{CKM}} \geq 3$ to $N_{\text{CKM}} = 3$ cannot be another CKM phase-counting theorem.

Normal-form flavor cost

The upper-bound bridge may be represented by a normal-form cost functional

$$\mathcal{C}_{\text{flavor}}(N) = \mathcal{C}_{\text{fields}}(N) + \mathcal{C}_{\text{Yukawa}}(N) + \mathcal{C}_{\text{CP}}(N) + \mathcal{C}_{\text{stability}}(N) - \mathcal{R}_{\text{identity}}(N). \quad (14)$$

Equation (14) is a normal-form audit functional, not a fitted phenomenological model or a derived Standard

Model effective potential. Here $\mathcal{C}_{\text{fields}}$ represents additional chiral multiplets and anomaly/accounting burden; $\mathcal{C}_{\text{Yukawa}}$ represents additional Yukawa parameters and flavor-alignment burden; $\mathcal{C}_{\text{CP}}$ represents additional irreducible CP phases requiring calibration or stabilization; $\mathcal{C}_{\text{stability}}$ represents precision-electroweak, Higgs, flavor-changing-neutral-current, and cosmological constraints; and $\mathcal{R}_{\text{identity}}$ represents the functional benefit of crossing the first irreducible CP/T-orientation threshold. Each cost term grows nonlinearly with N : for the quark flavor sector alone, the number of physical masses, mixing angles, and CP phases scales as

$$2N + \frac{N(N-1)}{2} + \frac{(N-1)(N-2)}{2} = N^2 + 1,$$

so the jump from $N = 3$ to $N = 4$ increases the quark flavor parameter count from 10 to 17.

The minimality bridge is the conjecture that

$$N_{\text{min}} = \arg \min_{N \geq 3} \mathcal{C}_{\text{flavor}}(N). \quad (15)$$

If the functional benefit saturates after the first CP/T orientation and the costs grow with additional sequential chiral generations, the normal-form minimum occurs at $N = 3$. This minimality logic is adjacent to the P6 speed-resource ledger and the P7 invariant-protection framework, but neither is required for the CKM lower-bound theorem [3, 4].

Conjecture 1 (Minimal sufficient weak-flavor architecture). *Among CKM-participating sequential chiral flavor architectures satisfying the CP/T-orientation lower bound, the realized architecture minimizes a finite-system flavor cost of the form Eq. (14). Under monotone overhead beyond the first sufficient CP/T channel, the minimum occurs at $N_{\text{CKM}} = 3$.*

Fourth-generation caveat

The upper-bound bridge is challenged only by a sequential chiral generation participating in the same CKM-like weak charged-current architecture. Vector-like fermions, sterile states, or hidden-sector states may extend phenomenology without counting as $N_{\text{CKM}} > 3$ in the X2 minimality claim. This distinction is important because vector-like quark searches and sequential fourth-generation models are different physical cases [14–17].

LEPTON-SECTOR EXTENSION

For Dirac neutrinos with an $N \times N$ PMNS matrix, the same phase-counting logic gives

$$N_{\text{CP}}^{\text{PMNS}} = \frac{(N-1)(N-2)}{2}. \quad (16)$$

If neutrinos are Majorana particles, additional Majorana phases appear, but they do not affect oscillation CP violation in the same way as the Dirac phase.

Proposition 1 (PMNS extension, conditional). *If the lepton sector participates in the same CP/T-oriented identity-transformation requirement, and if the relevant observable channel is the Dirac PMNS phase, then X2 motivates a nonzero leptonic Dirac CP phase under the stated assumptions.*

This is an extension, not the core theorem. Current global fits show that the determination of δ_{CP} remains ordering-dependent and not yet a settled nonzero measurement, so the PMNS statement is treated as an extension rather than part of the CKM lower-bound theorem [23]. Current and future long-baseline experiments such as T2K, NOvA, DUNE, and Hyper-Kamiokande test leptonic CP structure with increasing sensitivity [18–22]. If high-precision data establish $\delta_{\text{CP}} = 0$ or π , the lepton-sector extension would be demoted. The CKM lower-bound theorem would remain intact.

WHAT X2 DOES NOT EXPLAIN

X2 explains only the minimal dimensional requirement for a CKM-type irreducible CP/T orientation. It does not explain:

1. why the gauge group is $SU(3) \times SU(2) \times U(1)$;
2. why fermion masses have their observed hierarchy;
3. why CKM angles have their measured values;
4. why the CKM phase has its numerical value;
5. why neutrino mixing is large;
6. why the baryon asymmetry has its observed magnitude (this requires additional Sakharov-condition physics [13]);
7. why no beyond-Standard-Model fermions have been observed;
8. why the full Yukawa sector has its detailed texture.

These are not defects of the lower-bound theorem. They are open problems outside the scope of X2.

ALGEBRAIC DEMONSTRATIONS

The figures are deterministic algebraic or normal-form demonstrations generated by `code/generate_results.py`. They are not empirical fits.

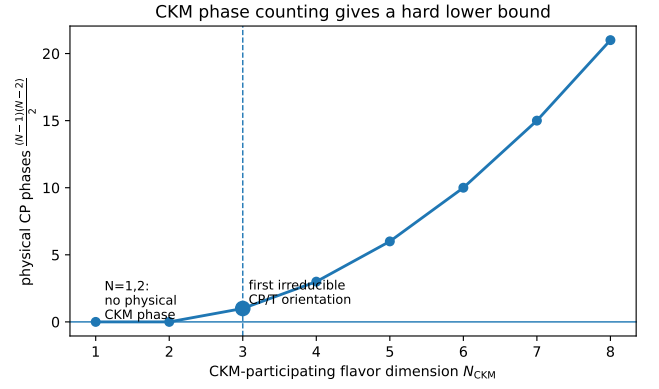


FIG. 1. Physical CKM-type CP phases as a function of CKM-participating flavor dimension. $N = 3$ is the first nonzero point. This is the hard algebraic hook of X2.

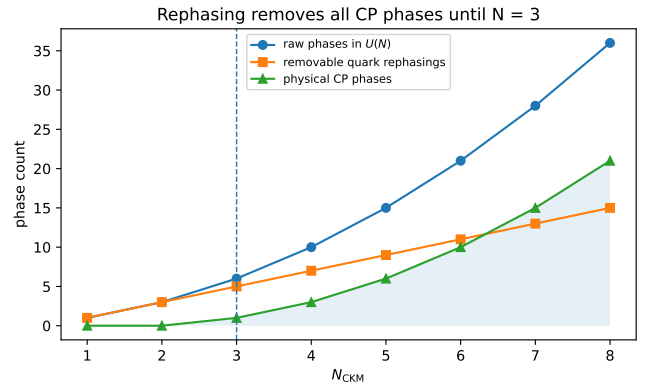


FIG. 2. Phase bookkeeping for an $N \times N$ unitary mixing matrix. Raw unitary phases grow as $N(N+1)/2$, but $2N-1$ quark-field rephasings are removable. The physical remainder is $(N-1)(N-2)/2$.

FALSIFICATION AND DEMOTION CONDITIONS

Lower-bound demotion

The lower-bound chain is weakened if any of the following occurs:

1. a CKM-like two-generation weak sector is shown to possess a physical irreducible CP phase;
2. the audited weak-sector identity-changing channel is fully realized without any CP/T orientation or compensating orientation variable;
3. the relevant identity-transformation carrier is not the CKM charged-current sector;
4. CPT assumptions fail in the relevant theory.

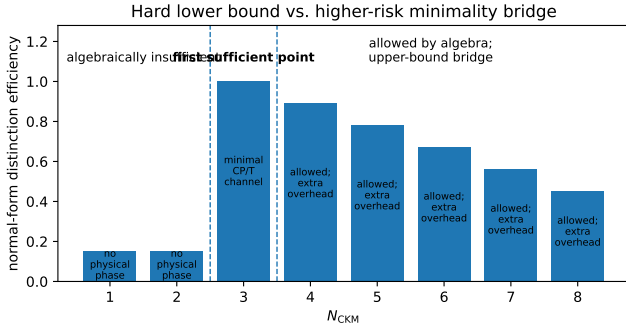


FIG. 3. Hard lower bound vs. higher-risk minimality bridge. $N = 1, 2$ cannot host an irreducible CKM CP phase. $N = 3$ is the first sufficient point. $N > 3$ is algebraically allowed but belongs to the upper-bound bridge.

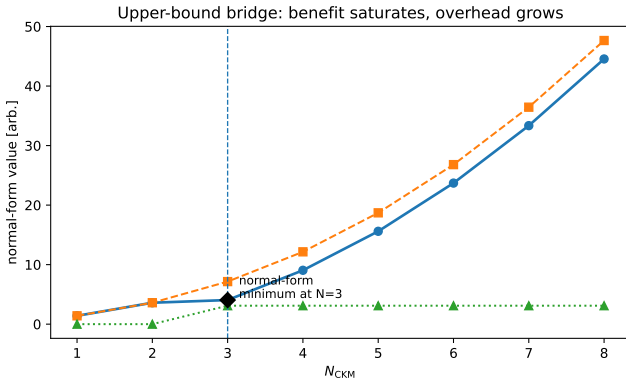


FIG. 4. Toy flavor-cost functional illustrating the upper-bound bridge. The functional benefit switches on at $N = 3$ and then saturates, while overhead continues to grow. This is a normal-form illustration, not a proof of the upper bound.

Upper-bound demotion

The minimality bridge is falsified or strongly demoted by the discovery of a fourth sequential chiral fermion generation with Standard Model gauge quantum numbers and weak charged-current participation. The lower bound $N_{\text{CKM}} \geq 3$ would survive. The $N = 3$ claim is also weakened if an $N > 3$ flavor architecture is shown to reduce the relevant cost functional or add a necessary nonredundant physical operation.

Leptonic extension demotion

The PMNS extension is demoted if high-precision oscillation measurements establish $\delta_{\text{CP}} = 0$ or π under assumptions requiring leptonic CP/T-oriented identity transformation.

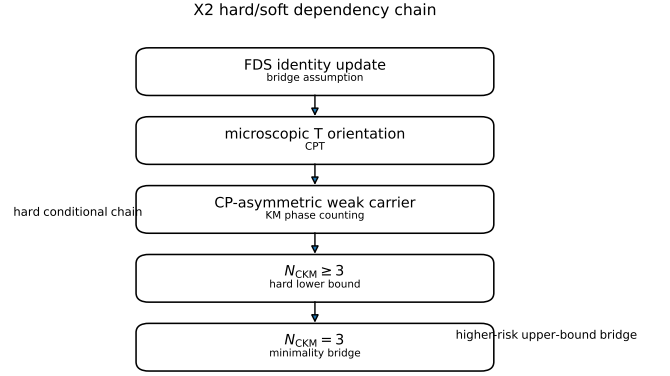


FIG. 5. X2 dependency chain. The hard conditional chain ends at $N_{\text{CKM}} \geq 3$. The final step to exactly three generations is a higher-risk minimality bridge.

RELATION TO EXISTING THEORY

X2 does not replace any of these theories. It packages one common implication inside FDS: if a finite-system weak identity transformation requires a microscopic CP/T orientation, the charged-current flavor architecture must have at least the dimensionality required for an irreducible CKM phase.

FUTURE WORK: MASS AS IDENTITY-SECTOR SEPARATION

The generation number is only one part of the flavor problem. A natural next question is whether the large fermion mass hierarchy encodes energetic separation barriers that prevent identity-sector ambiguity after the minimal CP/T-oriented architecture has been fixed. In such a program, masses would not merely be parameters but possible maintenance costs or separation scales for distinct identity sectors. X2 does not derive this; it marks it as a future problem.

CONCLUSION

The number three enters X2 not as numerology but as the first dimension in which weak flavor mixing can carry an irreducible CP/T orientation. A one-generation weak sector has no mixing phase. A two-generation weak sector has a Cabibbo angle but no physical CP phase. A three-generation weak sector has exactly one irreducible CKM phase, the minimal algebraic structure for CKM-type CP/T-oriented weak identity transformation.

The hard result is conditional but sharp:

$$\text{weak identity update} \Rightarrow T/CP \text{ orientation} \Rightarrow \text{irreducible CKM} \quad (17)$$

TABLE III. Relation of X2 to existing theory.

Existing theory	What it provides	X2 use
Kobayashi-Maskawa theory	CKM phase-counting result	Algebraic lower bound $N_{\text{CKM}} \geq 3$ [5].
CPT theorem	T violation linked to CP violation under QFT assumptions	Bridge from time-asymmetry to CP asymmetry [8, 9].
Standard Model flavor physics	Weak charged-current mixing and CKM phenomenology	Object-level carrier of identity transformation [7].
Jarlskog invariant	Rephasing-invariant CP-violation measure	Distinguishes physical phases from removable conventions [6].
Kaon and B-meson experiments	Observed quark-sector CP violation	Empirical anchor [10–12].
PMNS phenomenology	Possible leptonic CP violation and global-fit constraints	Conditional extension and future test [18–23].
Fourth-generation and vector-like fermion phenomenology	Distinguishes sequential chiral extensions from vector-like or hidden-sector variants	Clarifies what would and would not challenge the X2 upper-bound bridge [14–17].
FDS/DT	Identity-update and finite-system minimality language	Meta-level bridge selecting need for CP/T-oriented weak identity channel [1, 2].

The stronger statement $N_{\text{CKM}} = 3$, rather than $N_{\text{CKM}} > 3$, is a separate minimality bridge. Four or more CKM-participating generations are algebraically possible, but they are not required by the lower-bound theorem and must justify their additional flavor-sector cost. X2 should therefore be read as a conditional lower-bound theorem plus a quarantined upper-bound bridge.

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